



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC1001
Course title (English)	Introduction to Computer Science: Programming Methodology
Course title (Chinese)	计算机科学导论：程序设计方法
Units	3
Language of Instruction	English
Description (English)	This course introduces the basics of computer programming using Python. Students will learn the basic elements of modern computer systems, key programming concepts, problem solving and basic algorithm design. The key topics include the basic Python language syntax, data types, operators, flow control, defining and using function, I/O, data structure and algorithms, and the basics of object oriented programing. This course provides a foundation to further study in advanced computing topics.
Description (Chinese)	此课程基于 Python 编程语言讲授计算机编程基础。学生将学习到现代计算机系统的基本组成，重要的编程概念，解决问题的方法和基本的算法设计。本课程的主要内容包括了 Python 语言的基本语法，数据类型，运算符，程序流控制，定义和使用函数，输入输出，数据结构和算法，以及面向对象编程的基础知识。此课程将为以后学习进阶的计算课程打下基础。

2. Prerequisites / Co-requisites

A. Prerequisites

This course assumes no prior exposure to computer programming.

B. Co-requisites

N.A.

3. Learning Outcomes



Knowledge

- a) Students will understand the key elements of modern computer systems;
- b) Students will understand some key concepts and principles of modern computer programming languages;
- c) Students will understand basic data structures such as lists;
- d) Students will understand basic concepts of object oriented programming, such as object and class;
- e) Students will understand basic computer algorithms such as sorting and searching;

Skills

Generic:

- f) Student will be able to write clear and well-documented computer code in Python;
- g) Student will be able to write computer code that implements input/output;

Valued/Attitude:

- h) Students will appreciate the importance of computer science in real life;
- i) Students will be equipped with some basic knowledge of computer science, and be able to communicate with others concerning matters in computer science;
- j) Students will have awareness of the impact of computer science in social, industrial, environmental and technological context.

4. Course syllabus

This course will introduce the key programming concepts using Python language as examples, and explain the basics of algorithm design, using sorting and searching as examples. Some basic concepts of object oriented programming will also be covered in this course. This course will form a sound basis for further study of computer science.

The main topics covered in this course are:

- Introduction to modern computers;
- Preliminary knowledge for computer programming;
- Basic introduction to Python language;
- Data types and operators in Python language;
- Input/output;
- Flow control and loop;



- Function;
- List;
- Basic data structure;
- Introduction to algorithm design;
- Introduction to object oriented programming;

The mark for an assessment item submitted after the designated time on the due date, without an approved extension of time, will be reduced by 10% of the possible maximum mark for that assessment item for each day or part day that the assessment item is late. Note: this applies equally to week and weekend days.

5. Assessment Scheme

Component/ method	% weight
Assignments × 4	40
Mid-term quiz	20
Final exam	40

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to apply the principles in solving problems in novel situations, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented.



	Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

- A course questionnaire answered by every student after finishing all the lectures
- Reflections from teachers of more advanced courses
- Feedback from informal conversation over (free) coffee in my office and e-mail correspondence with students after class

8. Reading

9. Course components

Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
00000	Introduction to modern computers;
00001	Preliminary knowledge for computer programming
00000	Basic introduction to Python language;
00002	Data types and operators in Python language; Input/output
00000	Flow control and loop
00003	
00000	Function
00004	
00000	List
00005	
00000	Introduction to object oriented programming, part I



00006	
00000 00007	Review for mid-term quiz
00000 00008	Introduction to object oriented programming, part II
00000 00009	Data structure, part I
00000 00010	Data structure, part II
00000 00011	Introduction to algorithm design, part I
00000 00012	Introduction to algorithm design, part II
00000 00013	Introduction to algorithm design, part III
00000 00014	Review for final exam

11. Any other information

12. Version date

Version number	2
As of (date)	2023/07/18